fCAT: Assessing gene set completeness using domain-architecture aware targeted ortholog searches

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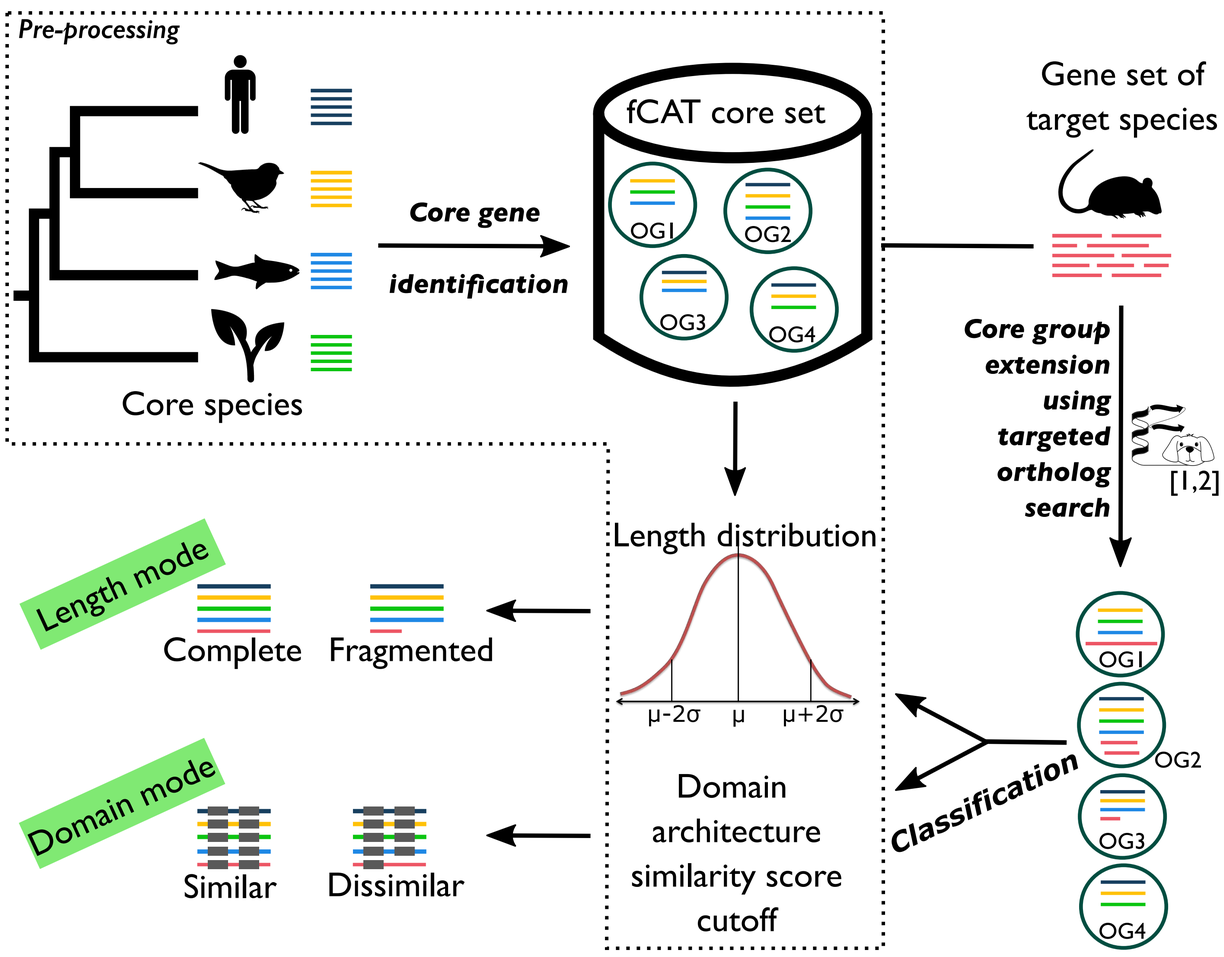
Motivation

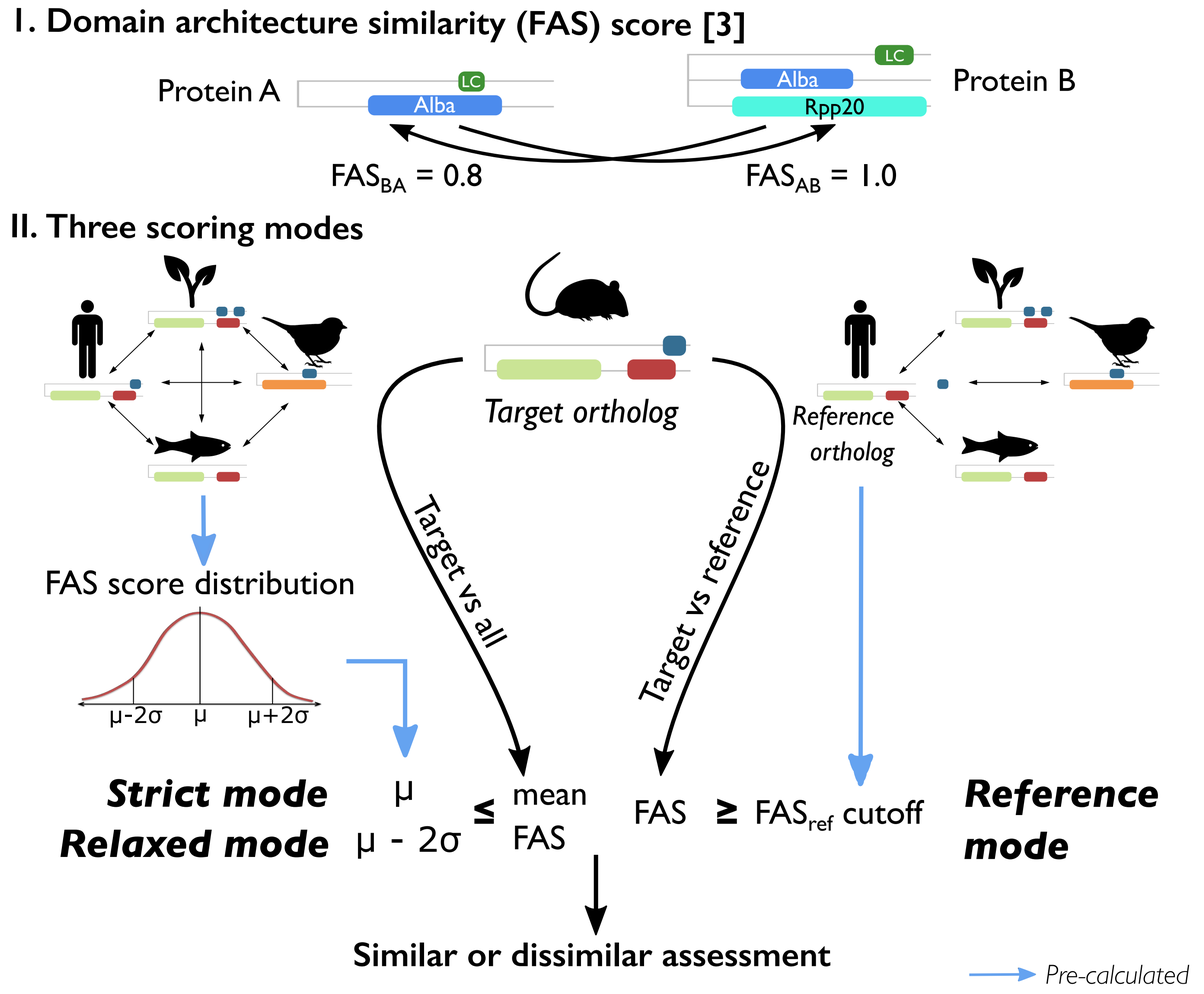
The assessment of gene set completeness is a routine task in genome analysis. Many common approaches start with identification of a set of single copy core genes for the taxonomic group the newly sequences species, the target, is part of. The fraction of missing core genes together with the extents of length deviation of the represented genes serve as a proxy for assessing the completeness of the target's gene set. Although well-established, this approach brings along a number of limitations: (i) typically, only single copy from the reference core genes are considered, (ii) often, only an error-prone unidirectional ortholog search is performed, and (iii) a simplistic length cut-off is applied to identify faulty gene predictions. Here we present fCAT, a novel algorithm using a targeted and domain-aware ortholog search for assessing gene set completeness of a target genome assembly. fCAT has three main innovations. First, it replaces the unidirectional search for core genes using a highly specific targeted ortholog search. Second, fCAT can consider differences in protein domain architectures to identify genes that most likely have been wrongly annotated. Third, results from the analysis of several species including the core species can be put into one context. This allows for comparative analyses rendering the differentiation between gene prediction artifacts and evolutionary events straightforward.

Method

fCAT WORKFLOW



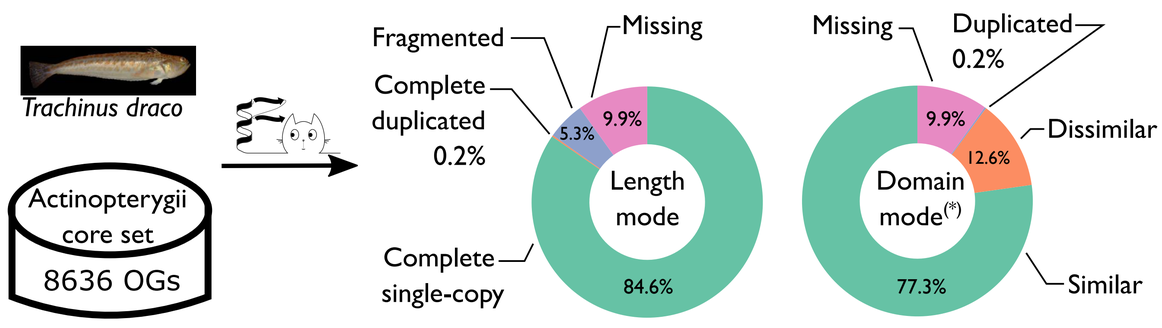
DOMAIN MODE



fCAT - Key Features

I. Basic completeness assessment

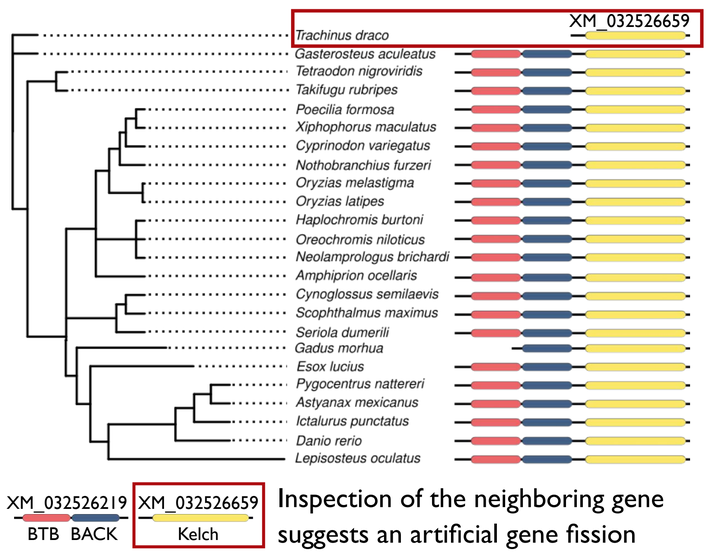
Statistic assessment for an input gene set of the target species



II. Effect of domain mode

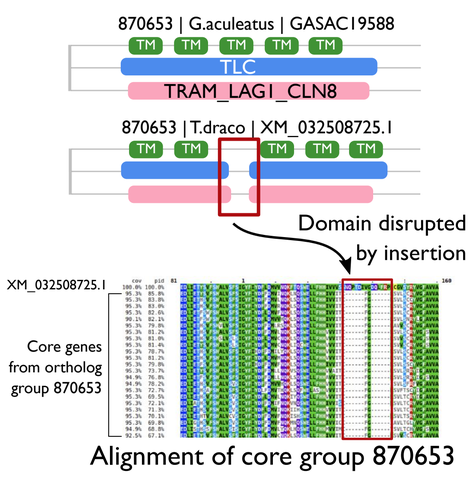
Comparing protein domain architectures between target ortholog and core genes provides more insight into the accuracy of gene annotation

Example 1: Artificial gene fission detected from fragmented classification



Inspection of the neighboring gene suggests an artificial gene fission

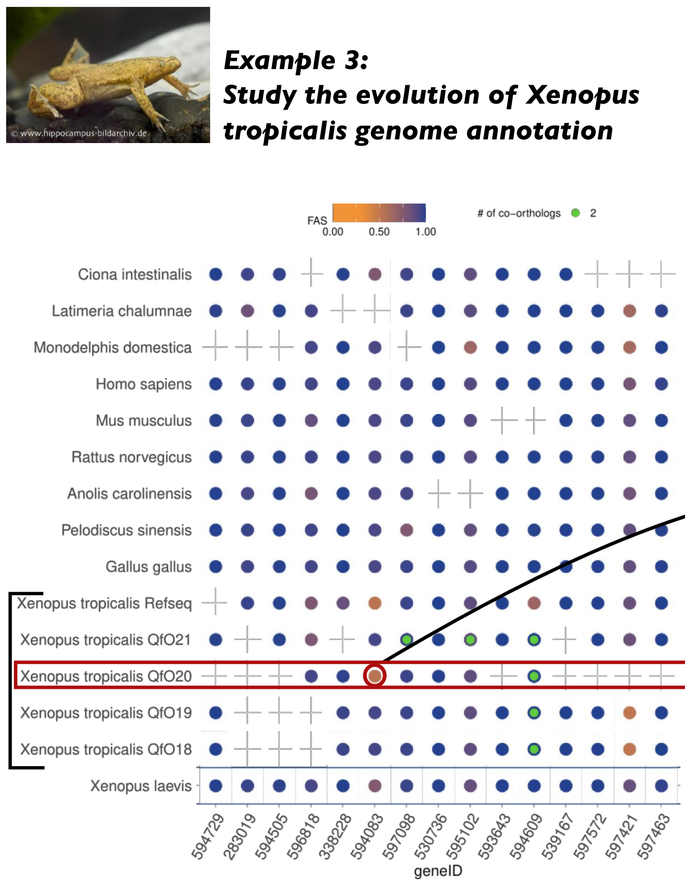
Example 2: Domain disruption due to erroneous gene annotation



Alignment of core group 870653 & *T.draco* ortholog

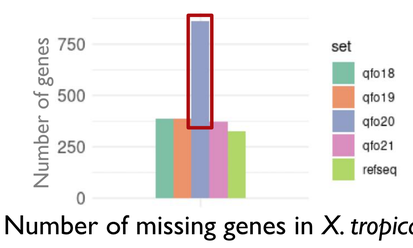
III. Informative phylogenetic profile output

Comparative study based on the phylogenetic profiles of the complete core genes across the core taxa and multiple target gene sets reveals the differentiation between gene prediction artifacts and evolutionary events



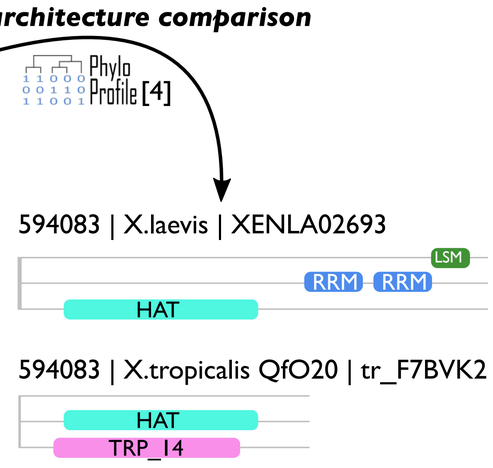
Phylogenetic profile of some core genes in Sarcopterygii core taxa and different *X. tropicalis* gene sets

Example 3: Study the evolution of Xenopus tropicalis genome annotation



Number of missing genes in *X. tropicalis* in different annotated gene sets

Protein domain architecture comparison



Contact

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References

[1] Ebersberger et al. (2009), BMC Evol Biol., 9:157
[2] https://github.com/BIONF/fDOG
[3] Dosch et al. (2022), bioRxiv, DOI: 10.1101/2022.09.01.506207
[4] Tran et al. (2018), Bioinformatics, 34(17):3041-3043

Availability

https://github.com/BIONF/fCAT
https://pypi.org/project/fcat/